Hidden in Plain Sight: Dog Whistle Detection, Classification, and Explanation

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Abstract

We study the detection, ingroup categorisation, and structured explanation of coded *dog-whistles* on Reddit, using the `silent_signals` corpus (Kruk et al. (2024)). Three open-weight systems are evaluated end-to-end. **(RQ-A)** A `RoBERTa-base` binary disambiguator trained on same-term coded/literal pairs reaches F1 = 0.707 ± 0.015 on a held-out human-gold disambiguation set (124 rows; F1 sits at the prevalence floor (0.701, the always-predict-coded baseline), so the meaningful improvement appears in accuracy (+15.6 pp) and PR-AUC (+0.30)) and performs equally when the curated glossary definition is withheld at inference. **(RQ-B)** A 17-way ingroup classifier scores macro-F1 = 0.353 ± 0.007 under a root-stratified split but 0.996 on a leaky alt-split gap that is the central methodological finding, attributable to a structural property of the underlying glossary (every term maps to exactly one ingroup; 0/696 ambiguous in train). **(RQ-C)** A `Flan-T5-XL`+LoRA generator emits parseable JSON 97% of the time and reaches `ingroup` macro-F1 = 0.379 ± 0.014; a class-balanced variant lifts headline accuracy by +12 pp while leaving macro-F1 unchanged (Δ = +0.004) and quadrupling seed variance, indicating reweighting rather than learned generalisation. Code, manifests, and per-row predictions are released.

1 Introduction

A dog-whistle is a phrase whose surface meaning is innocuous but which signals an in-group ideological position to those who recognise it (Henderson & McCready (2018)). Examples range from the surface-ambiguous (`absent fathers`, `globalist`, `urban`) to the conventionally pejorative (`SJW`, `libtard`). This deniability defeats standard toxicity classifiers and is actively exploited in extremist online discourse (Mendelsohn et al. (2023)). Automatic disambiguation, ingroup attribution, and human-readable rationale generation would all be valuable for moderation, auditing, and discourse research.

We address three tasks on the `silent_signals` Reddit corpus (Kruk et al. (2024)) (∼13 k coded examples, 298 root terms, 17 ingroup categories): **RQ-A** binary disambiguation of coded vs. literal use; **RQ-B** multiclass ingroup attribution; **RQ-C** structured-JSON generation of root, ingroup, definition, and explanation.

Three contributions follow. First, a binary disambiguator with a two-arm input ablation showing the curated glossary is *not* required at inference, a positive deployment-robustness finding. Second, we document a structural property of the Allen AI glossary (every term maps to exactly one ingroup) and show empirically that this renders ingroup classification trivially solvable on random splits but substantially harder under root-stratified evaluation; we argue root-stratification should be the default protocol on this corpus. Third, a structured-JSON generator that doubles as a negative result on class balancing: oversampling improves headline accuracy under prior shift but does not improve macro-F1 and quadruples seed variance.

2 Related Work

Dog-whistle detection. Henderson & McCready (2018) formalise dog-whistles as utterances that simultaneously address two audiences with different semantic content. Mendelsohn et al. (2023) released the Allen AI Glossary of Dogwhistles (∼300 phrases tagged with ingroup and definition), the direct source of our term vocabulary. Kruk et al. (2024) built `silent_signals` on this glossary via GPT-4 disambiguation of Reddit and Congressional text (85.3% precision on a 400-row human-validated sample); we use the informal (Reddit) split throughout.

Sense disambiguation, hate speech, and rationales. Our binary disambiguator frames each example as contextual word-sense disambiguation (Loureiro et al. (2021); Bevilacqua & Navigli (2020)): a target phrase plus context yields a sense label. The ingroup task echoes fine-grained hate-speech taxonomies (Vidgen et al. (2021); Mathew et al. (2021); Sap et al. (2020)), with the distinction that dog-whistles lack the surface offensiveness those systems rely on. For RQ-C we use `Flan-T5-XL` (Chung et al. (2022)) with LoRA adapters (Hu et al. (2022)), which keeps generation within a single A100 MIG slice while providing reliable structured output. Mathew et al. (2021) is the closest precedent for pairing classification labels with free-text rationales.

Methodology. We follow Bender & Friedman (2018) for the data statement (Appendix A), Mieskes (2017) for reproducibility hygiene, and Nozza et al. (2024) for report structure.

3 Data

Source corpus and filtering. `silent_signals` (Kruk et al. (2024)) contains ∼16 k dog-whistle occurrences across 298 canonical *roots*. We restrict to the informal (Reddit) split, retaining ∼13 k coded-sense rows after deduplication. Full data statement (Bender & Friedman (2018)) in Appendix A.

Splitting. Splits are **grouped by** `dog_whistle_root`: every root is assigned to exactly one of {train, val, test}. This forces the model to *generalise* to unseen phrases rather than memorise a glossary. The same partition is shared across RQ-A, RQ-B, and RQ-C. After grouping (seed 42, frozen): 219 train roots / 39 val / 40 test, zero overlap.

Negatives for RQ-A. `silent_signals` contains only positive (coded) occurrences. We mine literal-sense negatives in three stages: (i) heuristic keyword search over the SALT-NLP `informal_potential_dogwhistles` pool (∼6 M Reddit comments); (ii) LLM adjudication with `Llama-3.1-8B-Instruct` to drop coded-sense leakage; (iii) per-root 1:1 balancing. Stage 2's adjudication report flags a **62.11% contamination rate** on raw heuristic candidates – evidence the surface-form heuristic alone is unreliable. Train/val/test sizes after balancing: 12,366 / 2,522 / 2,768. Negatives are *silver*-labelled (LLM-as-judge); we bound the resulting ceiling in Appendix D. Pipeline details in Appendix B.

Locked human-gold evaluation (RQ-A only). Two files released with `silent_signals` are held entirely outside the training pipeline: `eval_detection.parquet` (n=101) and `eval_disambiguation.parquet` (n=124), both fully human-annotated. They are touched only by `model.evaluate()` after model selection on val. The 124-row disambiguation set is our headline RQ-A surface; its 54% coded prevalence yields a majority-class F1 floor of 0.701 (always-predict-coded saturation point) and an accuracy floor of 0.540.

The deterministic-glossary property. Of 696 distinct dog-whistle terms in train, **zero** map to more than one ingroup. The glossary, and the corpus that inherits from it, treats *term→ingroup* as a one-to-one function. We return to this in §6.

4 Methods

Architectures. For RQ-A and RQ-B we fine-tune `FacebookAI/roberta-base` (Liu et al. (2019)) with a single classification head. For RQ-C we use `google/flan-t5-xl` (Chung et al. (2022)) with **LoRA adapters** (Hu et al. (2022)) of rank 16 on the query and value projections of every transformer block; only the ∼36 MB adapter is trained, the base model is frozen. We initially attempted `microsoft/deberta-v3-large` (He et al. (2023)) for RQ-A and RQ-B but encountered numerical instability across five hyperparameter configurations on the cluster's `transformers` 5.x stack. We swapped to `RoBERTa-base`, which likely costs 1–3 F1 points – a small cost relative to seed variance on the 124-row eval set, and `RoBERTa-base` is the most-cited encoder in dog-whistle and hate-speech literature, making our numbers directly comparable to prior work.[1]

Input framing. RQ-A uses two arms: `term` (term + text + binary question) and `term_enriched_def` (adds the Allen AI glossary definition), isolating whether the curated glossary is needed at inference. A "text-only vs +term" ablation would be uninformative since the term is always lexically present. **RQ-B** contrasts `term` (term + text) against `text_only` (text alone). **RQ-C** prompts the model to return a JSON object with four fields (`dog_whistle_root`, `ingroup`, `definition`, `explanation`); an inference-time `repair_json()` step recovers missing braces, lifting

[1] All experiments run on a single A100 80 GB MIG slice (`4g.40gb`) on the Bocconi HPC cluster. RQ-A: 6 runs in 2 h 03 m. RQ-B: 4 runs in 13 m. RQ-C: 7 h 23 m (Path A) and 13 h 06 m (Path B) per 3-seed sweep.

parse rate from 0 to 0.97.

Training. All RQ-A/B runs use AdamW with lr=2e−5, weight decay 0.01, 10% warmup, batch size 32, max length 512, label smoothing 0.1, and head re-init at σ=0.02. RQ-C uses lr=3e−4, batch size 4 with grad accumulation 4, BF16 mixed precision, gradient checkpointing, and greedy decoding. All runs train up to 5 epochs with early stopping on val macro-F1 (patience 2). Headline numbers are mean ± std over seeds $\{42, 123, 7\}$. Baselines: majority-coded (F1 = 0.701, acc = 0.540 on `eval_disambiguation_124`; the high F1 floor is what an always-predict-coded baseline would achieve) and balanced random (F1 = 0.50 on the silver test split) for RQ-A; majority-class and frequency-weighted random for RQ-B; random-7-class (acc 0.143) and modal-class on the test mixture (acc 0.344) for RQ-C.

Metrics. For RQ-A, F1 of the coded class is primary; PR-AUC (precision-recall area under the curve, threshold-free) is reported as a secondary metric. For RQ-B, macro-F1 is primary because we care about both common and rare ingroups equally; weighted-F1 and accuracy are dominated by majority classes under the test prior shift. For RQ-C, JSON parse rate gates everything else; categorical fields are scored as exact-match classification; free-form `explanation` is scored with TF-IDF cosine similarity (overlap-weighted vector match between predicted and gold rationale, on a 0–1 scale).

5 Results

5.1 RQ-A – Binary disambiguation

Metric	`term`	`term_def`	Naive
F1 *disamb-124 (primary)*	**.707±.015**	.702±.021	.701 *(floor)*
Acc *disamb-124*	.696±.010	.677±.020	.540
PR-AUC *disamb-124*	.802±.010	.783±.006	—
F1 *detect-101 (secondary)*	.721±.042	.744±.049	.671
F1 *silver test* (n=2,768)	.792±.010	.768±.015	.500

Table 1: RQ-A. Mean±std over seeds $\{42, 123, 7\}$. `term_def` = `term_enriched_def`. "Floor" = what an always-predict-coded baseline achieves.

The two arms are within 0.005 F1 on the headline surface – the curated glossary is not required at inference. Per-seed variance on the 124-row eval is tight (σ=.015); the smaller detection set has a wider band ($\sigma\approx$.045). Silver-test F1 inherits label noise from train (Appendix D).

5.2 RQ-B – Multiclass ingroup

Run	Split	Input	m-F1	w-F1	Acc
A	Grouped	term	**.353±.007**	.420	.478
B	Grouped	text-only	.314	.447	.493
C	Alt (rand.)	term	**.996**	.999	.999
D	Grouped	term + bal-CE	.335	.360	.450

Table 2: RQ-B. Run A is our reported result (3-seed mean±std), n=2,095 test rows. Run C uses an alternative split where roots are not held out (n=1,947, single seed); deliberately leaky control.

Per-class F1 on Run A (full breakdown in Table 7, Appendix) highlights two extremes: `racist` and `transphobic` perform well, while **`anti-liberal`** **.000±.000** (n=720, 34% of test) collapses entirely. The modest macro-F1 partly reflects genuine ideological overlap: anti-liberal and white-supremacist rhetoric share vocabulary and frequently co-occur, so when the model confuses these categories (524/720 anti-liberal rows predicted as white supremacist), it may be tracking real semantic overlap rather than making arbitrary errors.

5.3 RQ-C – Structured generation

Metric	Path A	Path B	Δ
`json_parse_rate`	.975±.017	.969±.022	flat
ingroup macro-F1	**.379±.014**	.383±.063	+.004
ingroup accuracy	.487±.021	.606±.065	+.119
ingroup weighted-F1	.431±.026	.608±.119	+.177
root macro-F1	.160±.007	.167±.014	+.007
def. macro-F1	.016±.003	.021±.003	+.005
expl. cosine (TF-IDF)	.510±.011	.542±.010	+.032

Table 3: RQ-C. Path A = unbalanced; Path B = oversampled to 1,500 rows per ingroup. Both beat the modal-class accuracy floor (0.344). n=2,095 test rows.

6 Analysis

6.1 The grouped-vs-alt-split contrast

Runs A and C share architecture, hyperparameters, input format, and dataset rows; the sole difference is whether each `dog_whistle_root` is held out from train. Because term→ingroup is a one-to-one function in the glossary (§3), the two splits make glossary lookup either trivial or impossible (Table 4):

Split	Test rows whose term appears in train
Grouped (Run A)	0 / 2,095 (0%)
Alt (Run C)	1,935 / 1,947 (99%)

Table 4: Test-time term overlap with train under each split.

In Run C, 99% of test terms appear in train (median 21 rows per `(term, ingroup)` combo); the task collapses to reading a lookup table, and 99.6% macro-F1 reflects the corpus's deterministic glossary, not the model. In Run A, every test term is novel; the model must use context – and the decrease (.996 → .353) exposes where context is insufficient: 720/720 `anti-liberal` test rows use unseen terms, and the signal does not transfer from held-in training phrases (`SJW`, `snowflake`, `social justice warrior`, . . .) to held-out ones; F1 = 0.000 is the honest outcome. Reporting Run C as the headline would be reporting glossary memorisation; **the gap *is* the result.** Any future ingroup-classification work on `silent_signals` that does not hold out roots is measuring the glossary; we recommend root-stratification as the default evaluation protocol.

6.2 RQ-A – Modest but honest

RQ-A is evaluated on three surfaces: the ordinary silver test, plus two locked human-gold subsets for disambiguation and detection. Performance is strongest on the silver test ($F1 = .792$) and lower on the locked subsets ($F1 = .707$ for disambiguation, $F1 = .721$ for detection), suggesting that the model learns useful coded-versus-literal signal but struggles more on context-sensitive cases.

The headline disambiguation $F1$ is close to the prevalence floor (.707 vs. .701), so the stronger evidence comes from accuracy (+15.6 pp over the majority baseline) and PR-AUC (+0.30), which shows better-than-random ranking of coded over literal cases. Adding the glossary definition changes the headline score by only $\Delta = .005$ $F1$, so the curated glossary does not appear necessary at inference time.

6.3 RQ-C – Path B is reweighting, not learning

Oversampling to 1,500 rows per ingroup ($\times 17 =$ 25,500 rows) lifts headline accuracy from .487 to .606 (+12 pp) and weighted-F1 from .431 to .608 (+18 pp). Both gains are dominated by the test-dominant `anti-liberal` class (34% of test, 5.8% of natural train, ×6 prior shift). The clearest evidence that this is reweighting rather than improved learning on rare classes is that **macro-F1 does not move** ($\Delta = +\,.004$, well inside the seed std), while seed variance grows 4.5× (σ .014 → .063, with seed 7 collapsing to .296 and seed 42 reaching .438). A genuinely better intervention would not produce this variance pattern. Path A is the recommended system; Path B is reported for transparency.

6.4 Structural ceilings on RQ-C

`root.macro_f1`≈ .16 and `definition.macro_f1`≈ .02 are not bugs. Under the grouped split, no test root appears in train, so the canonical root string cannot be regenerated except through Flan-T5's pretraining prior or approximate matching from context. `definition` is a deterministic function of `root`, so its accuracy is structurally bounded by root accuracy. The meaningful generalisation question lives at the `ingroup` level, where all seven test ingroups are represented in train (just for *other* roots). We mark `ingroup.macro_f1` as the primary RQ-C metric and disclose the bounded fields explicitly.

Per-root F1 breakdowns and confusion matrices in Appendix C.

7 Conclusion

We present a three-task pipeline for dog-whistle detection, ingroup categorisation, and structured explanation on Reddit. Headline results are F1 $= 0.707 \pm 0.015$ (RQ-A, locked human-gold disambiguation), macro-F1 $= 0.353 \pm 0.007$ vs 0.996 (RQ-B, grouped vs alt-split), and `ingroup` macro-F1 $=$ 0.379 ± 0.014 (RQ-C, structured JSON, 97% parse rate). The grouped-vs-alt-split contrast is the central methodological finding: under the corpus's deterministic term-to-ingroup glossary, ingroup classification is trivial when roots leak and substantially harder when they do not. We argue root-stratification should be the default evaluation protocol for any future work on this corpus. Future work: a TF-IDF 1-NN retrieval baseline against the corrected RQ-C scorer, a per-root accuracy breakdown for RQ-B to localise the `anti-liberal` collapse, and a calibration analysis for the RQ-A binary scores.

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Appendix

A Data Statement

Curation rationale	Keyword search over Reddit (2005–2022, 45 subreddits) for ∼1,000 surface forms from the Allen AI Glossary; ∼6 M candidates sampled to 100,000; GPT-4 ensemble disambiguation. Informal subset only.
Language variety	English (`en-US`), informal social-media register.
Speaker demographic	Reddit users; US-skewed; no individual metadata.
Annotator demographic	Labels generated by GPT-4 with 400-row human-validated sample (85.3% precision; SALT-NLP, Georgia Tech / Stanford).
Speech situation	Public, asynchronous, written; speakers unaware of research use.
Text characteristics	10–18,300 chars per comment; topics: US politics, race, gender, religion, immigration.
Known biases	Ingroup distribution heavily skewed; `anti-liberal` rare in train (5.8%) but 34% of test; content targets minority groups.

Table 5: Data statement (Bender & Friedman (2018)) for the curated subset of `silent_signals`.

B Negative Mining Pipeline

The pipeline runs in three stages.

Stage 1. We stream candidates from `informal_potential_dogwhistles` for each dog-whistle surface form, mining up to 3× the positive count per term (capped at 2,000) to give Stage 2 enough headroom to survive its contamination losses. Rows shorter than 50 characters are discarded, and a SHA-256 content-hash exclusion set ensures no known positive or locked-eval row enters the negative pool.

Stage 2. Each surviving candidate is passed to `Llama-3.1-8B-Instruct` with the same prompt Kruk et al. (2024) used (phrase, coded meaning, comment text, forced binary answer); anything labelled `coded` is discarded. Using an independent model family means the same annotator is not constructing both classes of training data.

Stage 3. For each surface form, exactly $\min(K, n_{\mathrm{clean}})$ negatives are matched to the same number of positives. Any term with fewer than 3 clean candidates is dropped entirely – positives included – because keeping all its positives against zero negatives would teach the model keyword detection rather than disambiguation. The drop costs 100 terms and 1,037 positives (∼10.5% of pre-drop total); nearly all are pejorative-only coinages (`(((echo)))`, `troons`, `YWNBAW`) with no literal contrast. Surviving negatives inherit the same grouped-by-root split assignment as their matched positives.

C Extended Error Analysis

RQ-A: Binary disambiguation

RQ-A has three evaluation surfaces: the ordinary silver test, the locked disambiguation subset, and the locked detection subset. The ordinary silver test is broader and closer to the training distribution; the locked subsets are smaller human-gold stress tests. Disambiguation asks whether a known expression is literal or coded in context, while detection asks whether the instance should be flagged as coded. The main pattern is that performance is stronger on the ordinary silver test and weaker on the locked subsets. This suggests that the model learns a real coded-versus-literal distinction, but that this distinction becomes less stable in ambiguous, context-heavy cases. The main error concern is false negatives: coded cases predicted as literal. These matter because they are missed dog-whistle uses. Performance also varies across dog-whistle roots, so the average RQ-A score hides uneven robustness across phrase families.

RQ-B: Ingroup confusion matrix

True \ Pred	anti-lib.	racist	antisem.	wh. sup.	transph.	Islamoph.	conserv.
anti-liberal	0	48	36	**524**	38	0	0
racist	0	**395**	3	21	2	0	1
antisemitic	22	21	**214**	3	8	25	0
white supremacist	0	23	85	**115**	0	0	50
transphobic	0	0	0	12	**244**	0	0
Islamophobic	0	0	58	0	0	**59**	0
conservative	0	4	0	2	0	1	**2**

Table 6: RQ-B confusion matrix (Run A, root-stratified, `format_term`, seed_123, n=2,095). Bold = correct predictions. `anti-liberal` (n=720, 34% of test) collapses entirely into `white supremacist` (524/720 rows); 0 correctly classified. `conservative` (n=9) effectively unseen.

Ingroup	F1	n
`racist`	.882±.013	422
`transphobic`	.689±.186	256
`Islamophobic`	.648±.105	117
`antisemitic`	.623±.005	298
`white supremacist`	.250±.029	273
`conservative`	.083±.020	9
`anti-liberal`	.000±.000	720

Table 7: RQ-B per-class F1, Run A (mean across seeds $\{42, 123, 7\}$).

RQ-C: Ingroup confusion matrix

True \ Pred	Islamoph.	anti-lib.	antisem.	conserv.	racist	transph.	wh. sup.
Islamophobic	**86**	0	29	0	2	0	0
anti-liberal	0	0	**671**	0	49	0	0
antisemitic	0	0	**225**	9	45	0	19
conservative	0	0	9	**0**	0	0	0
racist	0	0	0	0	**397**	0	22
transphobic	0	10	5	0	0	**146**	71
white supremacist	0	0	1	0	75	0	**130**

Table 8: RQ-C ingroup confusion matrix (Path A, seed_123). `anti-liberal` collapses into `antisemitic` (671/720), mirroring the RQ-B failure. `conservative` (n=9) never correctly predicted. `racist` and `Islamophobic` are the most reliable classes.

D Limitations and Ethics

Limitations

Silver-label ceiling. Train and silver-test negatives are adjudicated by `Llama-3.1-8B-Instruct`. Stage 2's 62.11% contamination flag rate is evidence the pipeline does useful work, but the residual error rate is unmeasured. Two known bias modes remain: (a) genuine coded uses surviving Stage 2 as `non-coded`, depressing recall on the locked human-gold set; (b) systematic LLM error on sarcasm, ingroup reclamation, and meta-commentary on slurs. Positives also carry noise: `silent_signals` was constructed by GPT-4 disambiguation at 85.3% precision (400-row manual sample) – approximately one in seven training positives may be a literal use mislabelled as coded. We expect the true (fully-human-annotated) ceiling to be higher than 0.707 by an unknown but non-zero margin. The locked disambiguation set has n=124; a Wilson 95% confidence interval at p=.707 spans [.62, .78].

Scope and representativeness. All training and silver-test data are Reddit, 2005–2022, English, US-skewed. We do not claim findings transfer to other platforms, languages, or political eras. The Allen AI glossary may under-represent communities and forms of coded language not on the compilers' radar at the time of construction. The locked sets mix formal (Congressional) and informal (Reddit) examples while our model trains on Reddit only; reported F1 is a blend of in-domain and out-of-domain performance, kept unfiltered so any degradation on formal examples is a reportable finding.

Ethics Statement

Reddit data is public but users did not consent to NLP research; we surface no usernames or post IDs. Training data was labelled by two distinct LLM annotators with no human involvement beyond a 400-row validation sample: GPT-4 for the positive (coded) examples in `silent_signals`, and `Llama-3.1-8B-Instruct` for the mined negative (literal) examples. Neither model is neutral; systematic bias in either annotator propagates directly into the supervised signal. A deployed dog-whistle detector could be misused in two directions: *over-censorship*, flagging legitimate uses of ambiguous terms; or *under-censorship*, providing false security while novel dog-whistles outside the glossary go undetected. We frame all results as research findings on a fixed vocabulary and time window, not deployment recommendations.